Multiplying & Factoring Quadratics

The Box and X Methods

Complete Unit · Lesson, Practice & Answer Key

Section I — Lesson

Part 1	Forward: multiplying two binomials with a box
Part 2	Where the X comes from
Part 3	Backward: factoring when $a = 1$
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Part 5	Does the order of the split matter?
Part 6	The three sign cases
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Part 8	Summary — the whole method on one page

Section II — Practice Worksheet

A–F	30 problems, three difficulty levels

Section III — Complete Answer Key

Name: _____ **Date:** _____

Section I · Lesson

From multiplying binomials to factoring any quadratic

Key Idea

The big idea. The area of a rectangle is (length) × (width). If each side is a *binomial*, split the rectangle into four smaller rectangles — one for each pair of terms. The four little areas add up to the answer.

We learn the box in **two directions**:

- **Forward** (Part 1): given the two sides, *multiply* to find the area.
- **Backward** (Parts 3–7): given the area, *factor* to find the two sides.

Learning the forward direction first is what makes the backward direction make sense.

Part 1 • Forward — Multiplying Two Binomials

1.1 Simple case: leading coefficient $= 1$

☞ **Worked Example**

Multiply $(x+4)(x+5)$.

	x	$+4$
x	x^2	$4x$
$+5$	$5x$	20

x and $+4$ across the top	first factor
x and $+5$ down the side	second factor
cell = (row) × (column)	area of each piece
$x^2 + 4x + 5x + 20$	add all four cells
$= x^2 + 9x + 20$	combine middle terms

Notice: the two x-terms, $4x$ and $5x$, combine into the middle term $9x$. They sit on the **"/" diagonal** of the box, while x^2 and 20 sit on the **"\" diagonal**. Remember this — it is the key to factoring later.

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1.2 General case: leading coefficient $\neq 1$

☞ Worked Example

Multiply $(2x+3)(5x+7)$.

	$2x$	$+3$
$5x$	$10x^2$	$15x$
$+7$	$14x$	21

$$\begin{aligned} &10x^2 + 15x + 14x + 21 && \text{add the four cells} \\ &= 10x^2 + 29x + 21 && \text{combine } 15x + 14x \end{aligned}$$

The box is just the distributive property done twice:

$$(2x+3)(5x+7) = (2x+3)(5x) + (2x+3)(7) = 10x^2 + 15x + 14x + 21.$$

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Part 2 • Where the X Comes From

☞ Worked Example

Multiply the general pair $(x+p)(x+q)$ and watch the four cells.

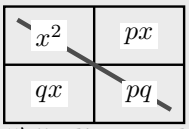
	x	$+p$
x	x^2	px
$+q$	qx	pq

$$(x+p)(x+q) = x^2 + \underbrace{(p+q)}_{\text{the } \textbf{sum}}\, x + \underbrace{pq}_{\text{the } \textbf{product}}$$

A 2×2 box has **two diagonals**. We name them by the way they slant:

Key Idea

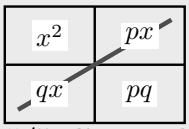
Naming the two diagonals.



"\" diagonal

holds x^2 and pq

the two you are **given**



"/" diagonal

holds px and qx

the two you must **find**

- The **"/" diagonal** *adds* to the middle term.
- The two diagonals have the *same product*: $x^2 \cdot pq = pq\,x^2$ and $(px)(qx) = pq\,x^2$.

House rule: always write the *variable* term first along the top and down the side. Then x^2 always lands in the top-left cell, and the **"\" diagonal** always carries the two known terms. *(If you write the terms in the other order the box still works — but the two diagonals swap places, so the names would flip.)*

☞ Worked Example

Now the fully general pair $(ax+b)(cx+d)$.

	ax	$+b$
cx	acx^2	bcx
$+d$	adx	bd

$$(ax+b)(cx+d) = acx^2 + (ad+bc)x + bd$$

$$\underbrace{(acx^2)(bd)}_{\text{\textbackslash\ diagonal}} = abcd\,x^2 = \underbrace{(bcx)(adx)}_{\text{/ diagonal}}$$

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Key Idea

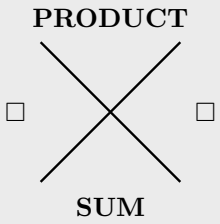
The equal-diagonals property. In *any* such box the two diagonals multiply to the same thing:

$$\underbrace{(acx^2)\cdot(bd)}_{\text{\textbackslash\ diagonal}} \;=\; \underbrace{(bcx)\cdot(adx)}_{\text{/ diagonal}} \;=\; abcd\,x^2.$$

Careful with words: all four cells sit in corners, and *both* pairs form a diagonal. Tell them apart by the **slant**: the **"\" diagonal** carries the x^2 term and the constant (what the problem gives you); the **"/" diagonal** carries the two x-terms (what you must find).

Key Idea

So every factoring problem asks the same two questions:



Top of the X = the product you need = $a \cdot c$

Bottom of the X = the sum you need = b

Sides of the X = the two numbers you are hunting for

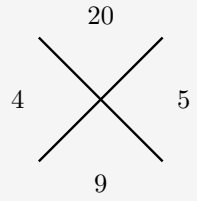
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Part 3 • Backward — Factoring when $a = 1$

☞ Worked Example

Factor $x^2 + 9x + 20$.

	x	$+4$
x	x^2	$4x$
$+5$	$5x$	20



top = product = 20
bottom = sum = 9
sides = the two numbers

Put x^2 and 20 on the \ diagonal	what we are given
need product 20, sum 9 $\Rightarrow$ 4 and 5	fill the / diagonal
read the sides: x+4 and x+5	GCF of each row/column

$$\boxed{x^2 + 9x + 20 = (x + 4)(x + 5)}$$

Key Idea

Shortcut when $a = 1$. You only need two numbers with **product** $= c$ and **sum** $= b$. Then $x^2 + bx + c = (x + p)(x + q)$ directly — no grouping required.

Part 4 • Backward — Factoring when $a \neq 1$

4.1 Why guessing is slow

To factor $6x^2 + 13x + 5$ by trial and error you would test every pairing:

Guess	Expands to	Middle term
$(x + 1)(6x + 5)$	$6x^2 + 5x + 6x + 5$	$11x$ ✗
$(6x + 1)(x + 5)$	$6x^2 + 30x + x + 5$	$31x$ ✗
$(2x + 1)(3x + 5)$	$6x^2 + 10x + 3x + 5$	$13x$ ✓

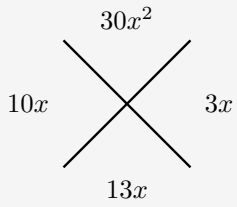
The X method finds the right one *directly*, with no guessing.

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4.2 The method: X, then split, then group

☞ Worked Example

Factor $6x^2 + 13x + 5$.



	$3x$	$+5$
$2x$	$6x^2$	$10x$
$+1$	$3x$	5

$ac = 6 \cdot 5 = 30$
need product 30, sum 13
$\Rightarrow$ 10 and 3

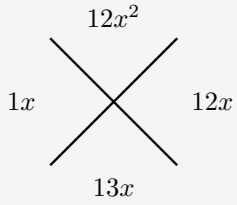
$$\begin{aligned} 6x^2 + 13x + 5 &= 6x^2 + 10x + 3x + 5 && \text{split the middle term} \\ &= 2x(3x + 5) + 1(3x + 5) && \text{GCF of each pair} \\ &= (2x + 1)(3x + 5) && \text{factor out the common } (3x + 5) \end{aligned}$$

Check: $(2x + 1)(3x + 5) = 6x^2 + 10x + 3x + 5 = 6x^2 + 13x + 5$ ✓

4.3 A second example — and the trap

☞ Worked Example

Factor $2x^2 + 13x + 6$.



$2x^2$	$1x$
$12x$	6

$ac = 2 \cdot 6 = 12$, sum 13
$\Rightarrow$ **1** and **12**

$$\begin{aligned} 2x^2 + 13x + 6 &= 2x^2 + 1x + 12x + 6 && \text{split} \\ &= \big(2x^2 + x\big) + \big(12x + 6\big) && \text{group in pairs} \\ &= x(2x + 1) + 6(2x + 1) && \text{GCF of each pair} \\ &= (2x + 1)(x + 6) && \text{brackets match } \checkmark \end{aligned}$$

✗ Common Mistakes

The trap. When you pull the GCF out of $2x^2 + x$, the leftover is $\mathbf{2x + 1}$, *not* $x + 2$:

$$2x^2 + x = x(2x + 1) \quad \checkmark \qquad\qquad 2x^2 + x \neq x(x + 2) \quad \boldsymbol{\times}$$

Writing $x(x+2)$ loses the leading coefficient and gives the wrong answer $(x+2)(x+6) = x^2+8x+12$ — a *different* polynomial.

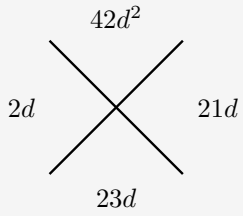
Divide, don't guess: $\dfrac{2x^2}{x} = 2x$ and $\dfrac{x}{x} = 1$, so the bracket must read $2x + 1$. If the two brackets do not come out **identical**, something is wrong — go back and check.

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Part 5 • Does the Order of the Split Matter?

☞ Worked Example

Factor $3d^2 + 23d + 14$ — and then do it again with the split reversed.



no common GCF · product $= a \cdot c = 3 \cdot 14 = 42$ · sum $= 23$

$42 = 2 \cdot 21$ and $2 + 21 = 23$ ✓ $\Rightarrow$ the numbers are **2** and **21**

First order ($2d$ then $21d$):

$$3d^2 + 23d + 14 = 3d^2 + 2d + 21d + 14 = d(3d + 2) + 7(3d + 2) = (3d + 2)(d + 7)$$

Second order ($21d$ then $2d$):

$$3d^2 + 23d + 14 = 3d^2 + 21d + 2d + 14 = 3d(d + 7) + 2(d + 7) = (d + 7)(3d + 2)$$

Check: $(3d + 2)(d + 7) = 3d^2 + 21d + 2d + 14 = 3d^2 + 23d + 14$ ✓

Key Idea

The order does not matter. Either split gives the same pair of factors — only the *order you discover them in* changes. Pick whichever grouping looks easier.

Finding the pair quickly. List the factor pairs of ac in order and test the sums:

Pair of 42	**Sum**	
$1 \cdot 42$	43	
$2 \cdot 21$	23	✓ **this one**
$3 \cdot 14$	17	
$6 \cdot 7$	13	

Going in order means you never miss a pair.

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Part 6 • The Three Sign Cases

Key Idea

Let the two mystery numbers be p and q, with $pq = ac$ and $p + q = b$.

Sign of ac	**Sign of b**	**What p and q look like**
$+$	$+$	both **positive**
$+$	$-$	both **negative**
$-$	either	**opposite signs**; the bigger one matches the sign of b

☞ Worked Example

Case $+/-$: Factor $6r^2 - 23r + 20$.

$ac = 6 \cdot 20 = 120 > 0$ and $b = -23 < 0$, so *both* numbers are negative: -15 and -8.

$$\begin{aligned} 6r^2 - 23r + 20 &= 6r^2 - 15r - 8r + 20 && \text{split} \\ &= 3r(2r - 5) - 4(2r - 5) && \text{pull out a } \textit{negative} \text{ GCF from the second pair} \\ &= (2r - 5)(3r - 4) && \text{brackets match } \checkmark \end{aligned}$$

The -4 is essential: taking $+4$ would give $4(-2r + 5)$, which does not match $(2r - 5)$.

☞ Worked Example

Case $-$: Factor $3n^2 + 2n - 8$.

$ac = 3 \cdot (-8) = -24 < 0$, so the numbers have *opposite* signs and sum to $+2$: $+6$ and -4.

$$\begin{aligned} 3n^2 + 2n - 8 &= 3n^2 + 6n - 4n - 8 && \text{split} \\ &= 3n(n + 2) - 4(n + 2) && \text{GCF of each pair} \\ &= (n + 2)(3n - 4) && \text{brackets match } \checkmark \end{aligned}$$

Part 7 • Always Take the GCF First

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☞ Worked Example

Factor $4x^2 + 14x + 6$.

Wrong start: jumping straight to $ac = 4 \cdot 6 = 24$ works, but the numbers get large and the final answer is not fully factored.

Right start: every term is even, so pull out 2 first.

$$4x^2 + 14x + 6 = 2\big(2x^2 + 7x + 3\big) \qquad \text{GCF} = 2$$

$$ac = 2 \cdot 3 = 6, \quad \text{sum } 7 \Rightarrow 6 \text{ and } 1 \qquad \text{much smaller numbers}$$

$$2x^2 + 7x + 3 = 2x^2 + 6x + x + 3 \qquad \text{split}$$

$$= 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3) \qquad \text{group}$$

$$\Rightarrow \; 4x^2 + 14x + 6 = \boxed{2(2x + 1)(x + 3)} \qquad \text{don't lose the 2!}$$

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Part 8 • Summary — The Whole Method on One Page

Key Idea

To MULTIPLY $(ax+b)(cx+d)$**:**

1. Write one factor across the top, the other down the side.
2. Fill each cell with (row) × (column).
3. Add all four cells and combine the two middle terms.

To FACTOR ax^2+bx+c **— the five steps:**

1. **GCF first.** Factor out anything common to all three terms.
2. **Multiply** $a \cdot c$. That is the **top** of the X.
3. **Find two numbers** with product ac and sum b. Those are the **sides** of the X.
4. **Split** the middle term into those two pieces, then **group in pairs**.
5. **Factor out the matching bracket** — and **check** by multiplying back.

When $a = 1$, Steps 2–4 collapse: just find two numbers with product c and sum b.

✗ Common Mistakes

- **Using** c **instead of** ac**.** When $a \neq 1$, the product is $a \cdot c$.
- **Losing the GCF.** If you factor out a 2 at the start, it must appear in the final answer.
- **Mismatched brackets.** $2x^2+x = x(2x+1)$, *not* $x(x+2)$. Divide each term by the GCF instead of guessing.
- **Wrong sign on the second GCF.** Sometimes you must pull out a *negative* to make the brackets match.
- **Mixing up the two diagonals.** ax^2 and c go on the **"\" diagonal**; the two x-terms go on the **"/" diagonal**. (Every cell is a corner — tell the diagonals apart by their slant.)
- **Not checking.** Multiplying back takes ten seconds and catches everything above.

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Section II · Practice Worksheet

30 problems · three difficulty levels

Key Idea

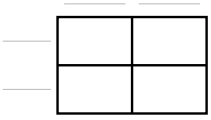
For each problem: write $a \cdot c$ on top of the X and b on the bottom, find the two numbers, split, group, factor — then **check by multiplying back**.

Difficulty: ●○○ Easy ●●○ Medium ●●● Hard

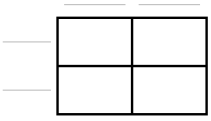
Practice A • Multiplying Binomials with a Box

Fill the box, then write the product in standard form.

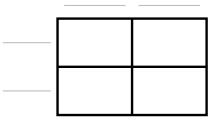
1. ●○○ **Easy** $(x+2)(x+6)$



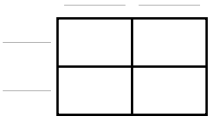
2. ●○○ **Easy** $(x+3)(x-5)$



3. ●●○ **Medium** $(2x+1)(x+4)$

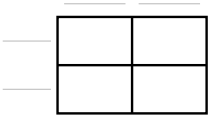


4. ●●○ **Medium** $(3x+2)(2x+5)$

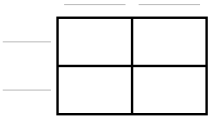


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5. ●●● **Hard** $(4x-3)(2x+7)$



6. ●●● **Hard** $(5x-2)(3x-4)$

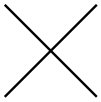


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Practice B • Factoring when $a = 1$

Two numbers with product c and sum b.

1. ●○○ **Easy** $x^2 + 7x + 12$



2. ●○○ **Easy** $x^2 + 10x + 21$



3. ●●○ **Medium** $x^2 - 9x + 20$



4. ●●○ **Medium** $x^2 + 2x - 15$



5. ●●○ **Medium** $x^2 - 3x - 28$



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6. ●●● **Hard** $x^2 - 16$ *(what is the middle term?)*

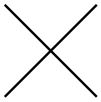


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Practice C • Factoring when $a \neq 1$ — All Signs Positive

Remember: the product is $a \cdot c$, not just c.

1. ●○○ **Easy** $2x^2 + 7x + 3$



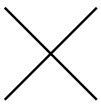
2. ●○○ **Easy** $3y^2 + 16y + 5$



3. ●●○ **Medium** $5m^2 + 17m + 6$



4. ●●○ **Medium** $4k^2 + 11k + 6$



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5. ●●○ **Medium** $4p^2 + 7p + 3$



6. ●●● **Hard** $6x^2 + 17x + 5$

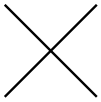


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Practice D • Working with Negative Signs

Check the sign of ac and of b before you start.

1. ●●○ **Medium** $2x^2 - 9x + 4$



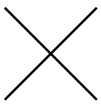
2. ●●○ **Medium** $5t^2 - 13t - 6$



3. ●●● **Hard** $4x^2 - 4x - 15$



4. ●●● **Hard** $6x^2 - 7x - 3$



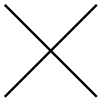
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Practice E • GCF First, Then Factor

Pull out the common factor before touching the X — and keep it in your answer.

1. ●●○ **Medium** $6y^2 + 21y + 9$



2. ●●○ **Medium** $8x^2 + 20x + 12$



3. ●●● **Hard** $12m^2 - 2m - 4$



4. ●●● **Hard** $18k^2 + 3k - 6$



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Practice F • Reasoning & Challenge

1. ●●○ **Medium** A rectangle has area $2x^2 + 11x + 12$ square units. One side measures $2x + 3$. Find the other side.

2. ●●○ **Medium** The product $12x^2 + bx + 15$ factors as $(3x + 5)$ times another binomial. Find the other binomial and the value of b.

3. ●●○ **Medium** Marcus factored $8x^2 + 10x - 3$ and got $(4x - 1)(2x + 3)$. Multiply his answer back out. Is he right? Explain how you know.

4. ●●● **Hard** Complete the box. Two cells are filled in; find the missing cells *and* both binomial factors.

$6x^2$	
	-10

Hint: there is more than one correct answer. Give one and check it.

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5. ●●● **Hard** Find *every* positive integer k for which $3x^2 + kx + 8$ factors with integer coefficients. *(Hint: $ac = 24$; list the factor pairs.)*

6. ●●● **Hard** **Why does $a \cdot c$ work?** Suppose $ax^2 + bx + c = (mx + p)(nx + q)$. Multiply the right side out and explain why the two numbers you hunt for must multiply to ac and add to b.

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Section III · Complete Answer Key

with the product–sum reasoning for every problem

Answer Key

[title=Practice A — Multiplying Binomials]

1. Cells $x^2,\ 6x,\ 2x,\ 12 \Rightarrow \mathbf{x^2 + 8x + 12}$
2. Cells $x^2,\ -5x,\ 3x,\ -15 \Rightarrow \mathbf{x^2 - 2x - 15}$
3. Cells $2x^2,\ 8x,\ x,\ 4 \Rightarrow \mathbf{2x^2 + 9x + 4}$
4. Cells $6x^2,\ 15x,\ 4x,\ 10 \Rightarrow \mathbf{6x^2 + 19x + 10}$
5. Cells $8x^2,\ 28x,\ -6x,\ -21 \Rightarrow \mathbf{8x^2 + 22x - 21}$
6. Cells $15x^2,\ -20x,\ -6x,\ 8 \Rightarrow \mathbf{15x^2 - 26x + 8}$

Answer Key

[title=Practice B — Factoring when $a = 1$]

1. Product 12, sum $7 \Rightarrow 3, 4$. $\quad \mathbf{(x + 3)(x + 4)}$
2. Product 21, sum $10 \Rightarrow 3, 7$. $\quad \mathbf{(x + 3)(x + 7)}$
3. Product 20, sum $-9 \Rightarrow -4, -5$ (both negative). $\quad \mathbf{(x - 4)(x - 5)}$
4. Product -15, sum $2 \Rightarrow 5, -3$ (opposite signs). $\quad \mathbf{(x + 5)(x - 3)}$
5. Product -28, sum $-3 \Rightarrow -7, 4$. $\quad \mathbf{(x - 7)(x + 4)}$
6. Write it as $x^2 + 0x - 16$: product -16, sum $0 \Rightarrow 4, -4$. $\quad \mathbf{(x - 4)(x + 4)}$ — a *difference of squares*.

Answer Key

[title=Practice C — Factoring when $a \neq 1$, All Positive]

1. $ac = 6$, sum $7 \Rightarrow 6, 1$: $\ 2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = \mathbf{(2x + 1)(x + 3)}$
2. $ac = 15$, sum $16 \Rightarrow 15, 1$: $\ 3y^2 + 15y + y + 5 = 3y(y + 5) + 1(y + 5) = \mathbf{(3y + 1)(y + 5)}$
3. $ac = 30$, sum $17 \Rightarrow 15, 2$: $\ 5m^2 + 15m + 2m + 6 = 5m(m + 3) + 2(m + 3) = \mathbf{(5m + 2)(m + 3)}$
4. $ac = 24$, sum $11 \Rightarrow 8, 3$: $\ 4k^2 + 8k + 3k + 6 = 4k(k + 2) + 3(k + 2) = \mathbf{(4k + 3)(k + 2)}$
5. $ac = 12$, sum $7 \Rightarrow 3, 4$: $\ 4p^2 + 3p + 4p + 3 = p(4p + 3) + 1(4p + 3) = \mathbf{(4p + 3)(p + 1)}$
6. $ac = 30$, sum $17 \Rightarrow 15, 2$: $\ 6x^2 + 15x + 2x + 5 = 3x(2x + 5) + 1(2x + 5) = \mathbf{(3x + 1)(2x + 5)}$

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Answer Key

[title=Practice D — Working with Negative Signs]

1. $ac = 8 > 0$, sum $-9 < 0 \Rightarrow$ both negative: $-8, -1$. $2x^2 - 8x - x + 4 = 2x(x-4) - 1(x-4) = \mathbf{(2x-1)(x-4)}$

2. $ac = -30 < 0$, sum -13: $-15, 2$. $5t^2 - 15t + 2t - 6 = 5t(t-3) + 2(t-3) = \mathbf{(5t+2)(t-3)}$

3. $ac = -60 < 0$, sum -4: $-10, 6$. $4x^2 - 10x + 6x - 15 = 2x(2x-5) + 3(2x-5) = \mathbf{(2x-5)(2x+3)}$

4. $ac = -18 < 0$, sum -7: $-9, 2$. $6x^2 - 9x + 2x - 3 = 3x(2x-3) + 1(2x-3) = \mathbf{(2x-3)(3x+1)}$

Answer Key

[title=Practice E — GCF First]

1. $= 3(2y^2 + 7y + 3)$; $ac = 6$, sum $7 \Rightarrow 6, 1$: $2y(y+3) + 1(y+3)$. $\mathbf{3(2y+1)(y+3)}$

2. $= 4(2x^2 + 5x + 3)$; $ac = 6$, sum $5 \Rightarrow 2, 3$: $2x^2 + 2x + 3x + 3 = 2x(x+1) + 3(x+1)$. $\mathbf{4(x+1)(2x+3)}$

3. $= 2(6m^2 - m - 2)$; $ac = -12$, sum $-1 \Rightarrow -4, 3$: $6m^2 - 4m + 3m - 2 = 2m(3m-2) + 1(3m-2)$. $\mathbf{2(3m-2)(2m+1)}$

4. $= 3(6k^2 + k - 2)$; $ac = -12$, sum $1 \Rightarrow 4, -3$: $6k^2 + 4k - 3k - 2 = 2k(3k+2) - 1(3k+2)$. $\mathbf{3(3k+2)(2k-1)}$

In every case the GCF stays in the final answer.

Answer Key

[title=Practice F — Reasoning & Challenge]

1. $ac = 2 \cdot 12 = 24$, sum $11 \Rightarrow 8, 3$; $2x^2 + 8x + 3x + 12 = 2x(x+4) + 3(x+4) = (2x+3)(x+4)$. Since one side is $2x + 3$, the other side is $\mathbf{x + 4}$. *Check:* $(2x+3)(x+4) = 2x^2 + 11x + 12$ ✓

2. The x^2 cells must give $12x^2$, so the other leading term is $4x$; the constants must give 15, so the other constant is 3. The other binomial is $\mathbf{4x + 3}$. Multiplying: $(3x+5)(4x+3) = 12x^2 + 9x + 20x + 15 = 12x^2 + 29x + 15$, so $\mathbf{b = 29}$.

3. $(4x-1)(2x+3) = 8x^2 + 12x - 2x - 3 = 8x^2 + 10x - 3$. **Marcus is correct.** ✓ *(Confirm with the X: $ac = 8 \cdot (-3) = -24$, sum $10 \Rightarrow 12$ and -2; $8x^2 + 12x - 2x - 3 = 4x(2x+3) - 1(2x+3) = (4x-1)(2x+3)$.)* Re-multiplying confirms a right answer just as clearly as it catches a wrong one.

4. With $6x^2$ and -10 on the **"\" diagonal**, $ac = 6 \cdot (-10) = -60$. Choose any pair with product -60 — for example 15 and -4 (sum 11):

$$6x^2 + 15x - 4x - 10 = 3x(2x+5) - 2(2x+5) = \mathbf{(3x-2)(2x+5)},$$

which expands to $6x^2 + 11x - 10$. *Other answers are possible* — each factor pair of -60 gives a different middle term.

5. $ac = 3 \cdot 8 = 24$. Each factor pair of 24 gives a value of k (their sum):

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Pair	k	**Factorization**
$1 \cdot 24$	25	$(3x+1)(x+8)$
$2 \cdot 12$	14	$(3x+2)(x+4)$
$3 \cdot 8$	11	$(3x+8)(x+1)$
$4 \cdot 6$	10	$(3x+4)(x+2)$

So $\mathbf{k = 10,\ 11,\ 14,\ 25}$.

6. Multiplying out,

$$(mx+p)(nx+q) = \underbrace{mn}_{a}x^2 + \underbrace{(mq+np)}_{b}x + \underbrace{pq}_{c}.$$

Now multiply the two middle pieces together:

$$(mq)(np) = (mn)(pq) = a \cdot c.$$

So the two terms that *add* to b are exactly the two terms that *multiply* to ac. That is precisely what the X asks for — and it is why splitting the middle term into those two pieces always makes the grouping work.

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